

REGRESSION: THE BASICS

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LEARNING OBJECTIVES

- Understand the steps for developing a regression model
- Understand and decipher regression assumptions
- Understand the differences in assumptions for linear and logistic regression

DATA TYPE DEFINITIONS

- **Continuous:** Quantitative variables that can any value within a given range, including decimal or fractions. It is measured, not counted.
 - Examples:
 - Height (176 cm)
 - Weight (85.5 kg)
 - Temperature (22.32 degrees)
- **Binary:** A categorical variable that can only take two possible values, usually yes/no, true/false, or 0/1.
- **Ordinal:** A categorical variable where the categories have a meaningful order/ranking, but the differences between categories are not numerically measured
 - Examples:
 - Customer Satisfaction: 1-5
 - Movie Rankings: 1-5 'Stars'
- **Count:** Numerical data that represents a number of times an event occurs. Non-negative whole numbers, and cannot be a fraction or decimal
 - Examples:
 - One fish, two fish, red fish, blue fish
 - Number of emails in a day

REGRESSION DEFINITIONS

- P-Value: Probability of observing a result greater than this.
- 95% Confidence Interval: If the analysis was performed 100 times you would expect the results to be within this bound 95 times.

REGRESSION IS THE 'SWISS ARMY KNIFE' OF ANALYSES

- Over 90% of analyses will be done with regression (of some sort)
- Extremely flexible, used across different data structures
 - Continuous
 - Count
 - Ordinal
- T-Test, ANOVA's, and even Deep Learning are regressions!

BASIC STEPS IN REGRESSION ANALYSES

- Identify your question to be answered
- Understand your data
- Identify the right analytical strategy
- Identify potential covariates (i.e., confounders) to include in the analysis
- Perform regression
- Assess assumptions through diagnostics
- Iterate until the analyses meet the assumption
- Infer conclusions

IDENTIFY YOUR QUESTION TO BE ANSWERED

- PICOT (Epidemiology lecture)
- Scientific model (DAG and confounding lecture)
- Are we assessing?
 - Associations
 - Prediction
 - Causality

Losciale JM, Bullock GS, Collins GS, Arundale AJ, Hughes T, Arden NK, Whittaker JL. Description, prediction, and causation in sport and exercise medicine research: resolving the confusion to improve research quality and patient outcomes. Journal of Orthopaedic & Sports Physical Therapy. 2023 Jul;53(7):381-7.

IDENTIFY YOUR QUESTION TO BE ANSWERED

TABLE 1		
TYPES OF RESEARCH		
Research Goal	Definition	Example(s)
Description ⁷	- The study of the existing distribution of factors among a target population or the reporting of how 2 distinct factors relate to one another. - There is no concern for a causal interpretation.	- A study sought to understand how frequent ice cream sales were during each summer month in a coastal town. They further assessed ice cream sales by flavor during the same months. - A study sought to understand if knee extension strength after a knee injury is related to walking biomechanics. They further assessed this association by gender.
Causation ⁷	- The study of the relation between exposure to a suspected cause and the possible effects they produce. - There can be more than one cause for any outcome. - A cause-effect relationship will be affected by confounding (TABLE 2).	- A study sought to understand the causal effect of exposure to a campfire (while camping) on hand skin burn injury incidence. The authors propose that exposure to a campfire will increase skin burn injury incidence among those camping compared to having not built a campfire while camping. - A study sought to understand the causal effect of a baseball pitcher's ER ROM on the incidence of medial elbow injuries. The authors propose that exposure to lower levels of ER ROM will increase medial elbow injury incidence among pitchers compared to having higher levels of ER ROM.
Prediction ²⁰	The process of developing and testing mathematical models (equations)^a that include multiple factors (obtained from descriptive and causal research) that forecast the risk/probability of an undiagnosed condition (diagnostic) or future outcome (prognostic)	See below

Losciale JM, Bullock GS, Collins GS, Arundale AJ, Hughes T, Arden NK, Whittaker JL. Description, prediction, and causation in sport and exercise medicine research: resolving the confusion to improve research quality and patient outcomes. Journal of Orthopaedic & Sports Physical Therapy. 2023 Jul;53(7):381-7.

UNDERSTAND YOUR DATA

- Identify the outcome
 - Continuous
 - Binary
 - Count
 - Ordinal
- Identify the exposure/explanatory variable
 - Continuous
 - Binary
 - Ordinal
- Identify other covariates and their data structure

IDENTIFY THE RIGHT ANALYTICAL STRATEGY

- Linear Regression: Continuous outcome
- Logistic Regression: Binary outcome
- Poisson Regression (not being discussed): Count outcome
- Ordinal Regression (not being discussed): Ordinal Outcome

IDENTIFY OTHER COVARIATES FOR INCLUSION

- Continuous
- Binary
- Ordinal
- Count

*Refer to 'DAG' module for scientific structure!

PERFORM REGRESSION

Time to bring back your basic algebra

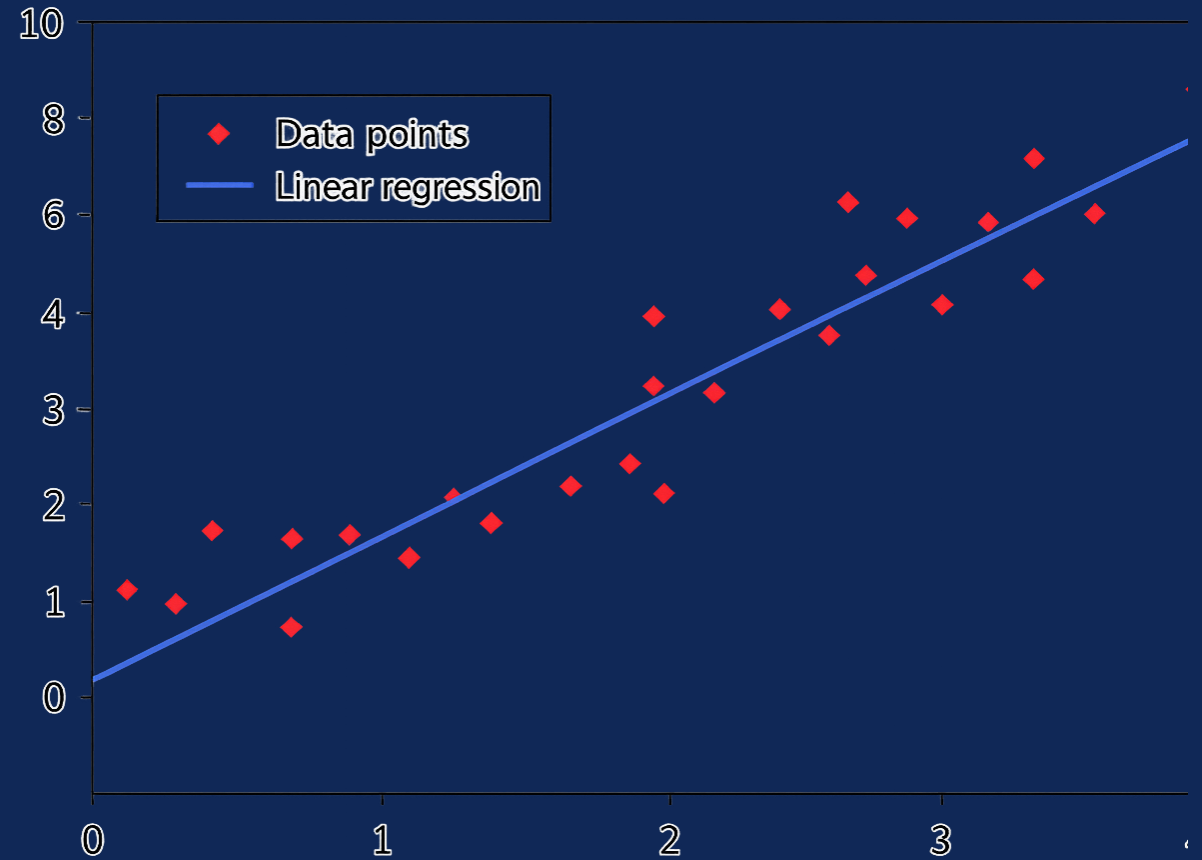
$$Y = \beta_0 + \beta_1 X + \epsilon$$

Y = Outcome

β_0 = Intercept (i.e., starting point)

$\beta_1 X$ = Coefficient (rise over run)

ϵ = error (uncertainty)



PERFORM REGRESSION

- Will start with 'basic' linear regression
- 'Basic' regressions are 'Base' R functions
- Will add different packages to help summarize the outputs

```
model <- lm(Y ~ X, data = data)  
summary(model)
```

Y = Outcome

X = Exposure

PERFORM REGRESSION

- I am not a big fan of the base 'lm' function
- Prefer to use glm function (still 'base' R)
- Will integrate the 'Epi' package to create a better regression output

```
library(Epi)
```

```
model <- glm(Y ~ X, data = data, family = 'gaussian')
```

```
ci.lin(model)
```

Y = Outcome

X = Exposure

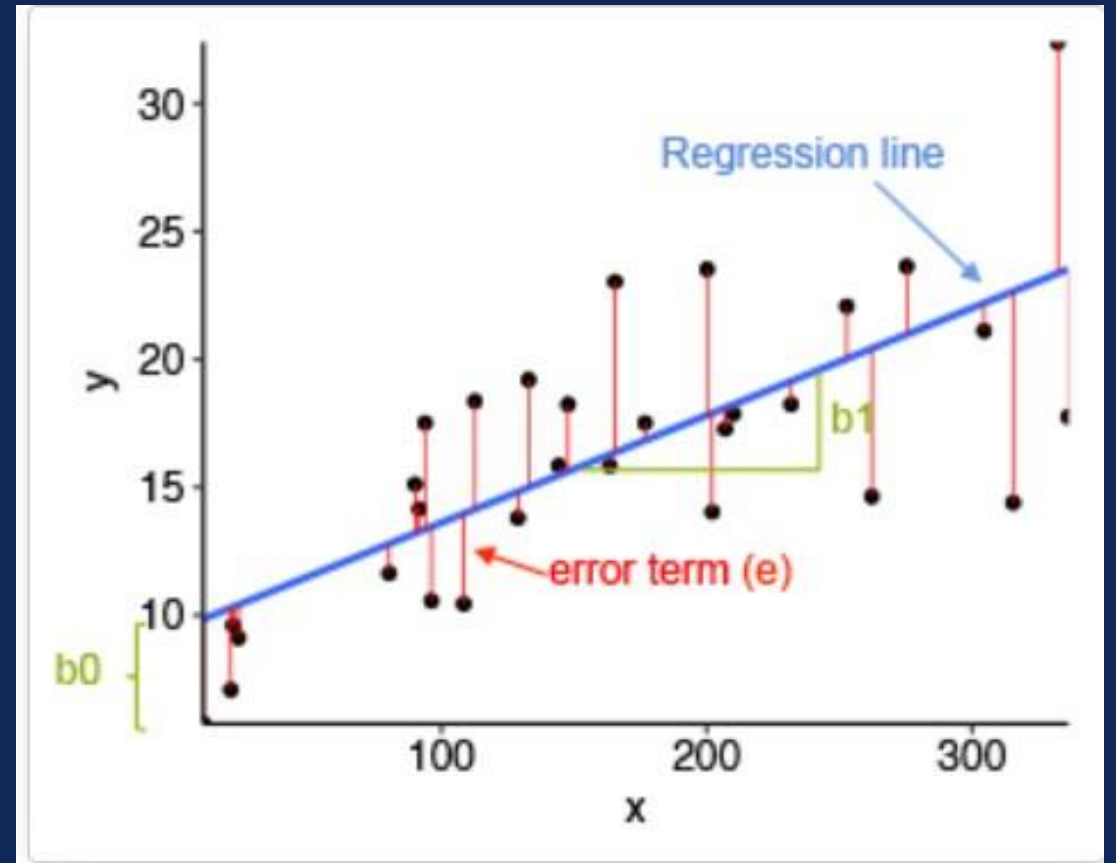
ASSESS ASSUMPTIONS

- Basic regression code as described before
- `model <- glm(Y ~ X, data = data, family = 'gaussian')`
- `summary(model)`

- Now to assess diagnostics for assumptions
- `plot(model)`

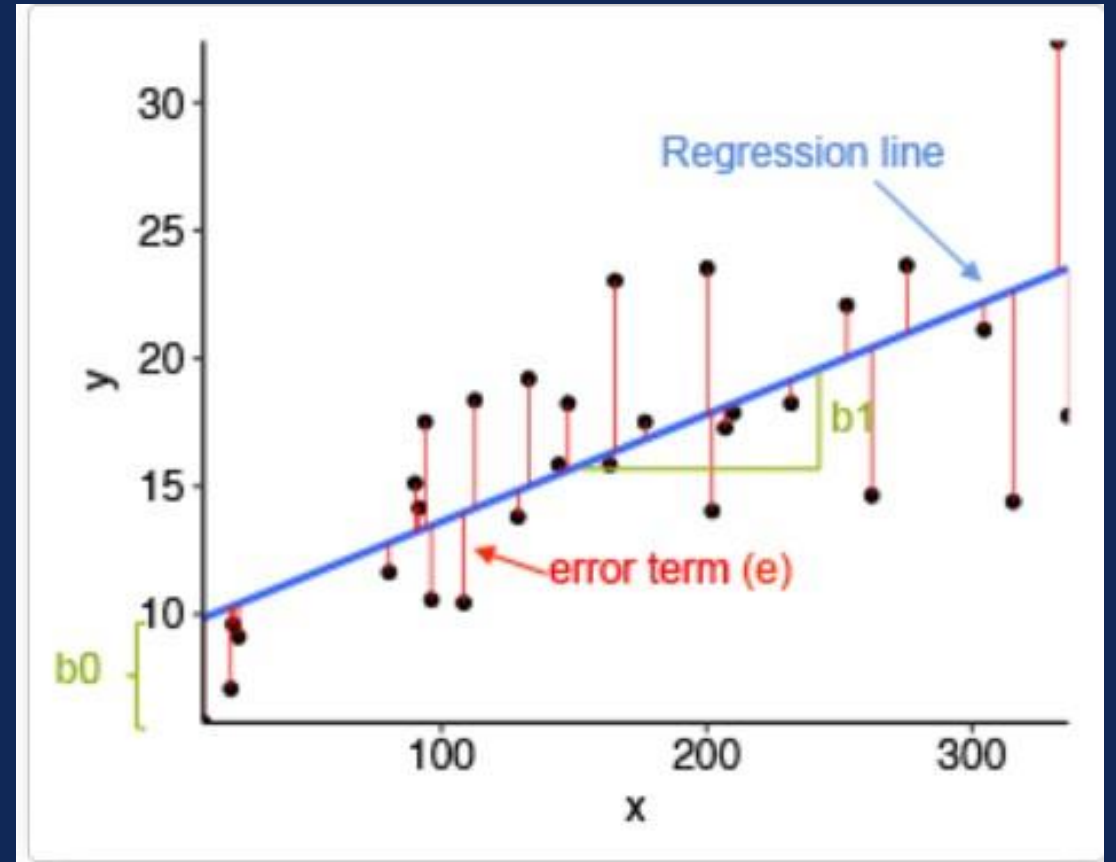
ASSESS ASSUMPTIONS

- This is key to create stable results.
- Assumptions are based on the RESIDUALS
- NOT raw data



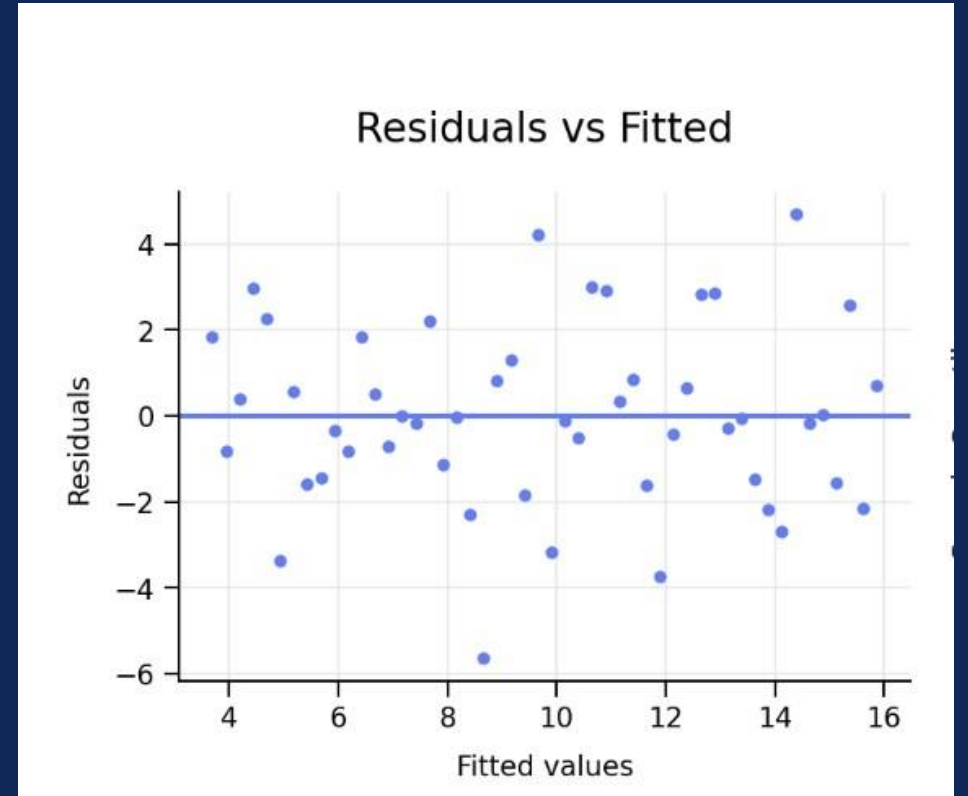
ASSESS ASSUMPTIONS

- Linearity
- Independence of observations
- Normality
- Constant (equal) variance



ASSESS ASSUMPTIONS

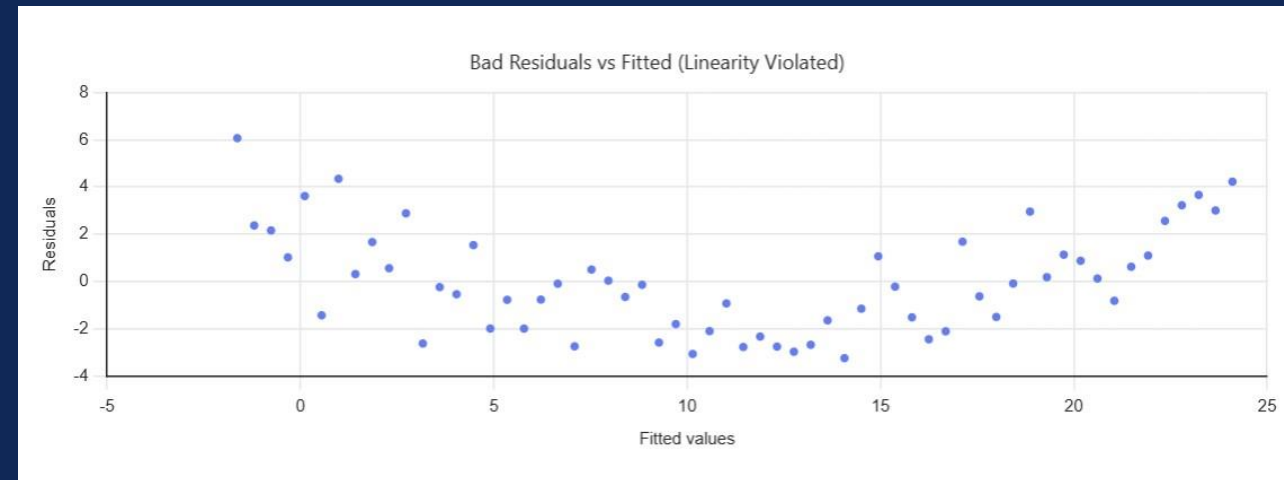
- Linearity: Association between factors and outcome are linear
- How to assess: Residuals versus fitted
- What are we looking for: No curvature



'Good' Plot

ASSESS ASSUMPTIONS

- Linearity: Association between factors and outcome are linear
- How to assess: Residuals versus fitted
- What are we looking for:
 - No curvature



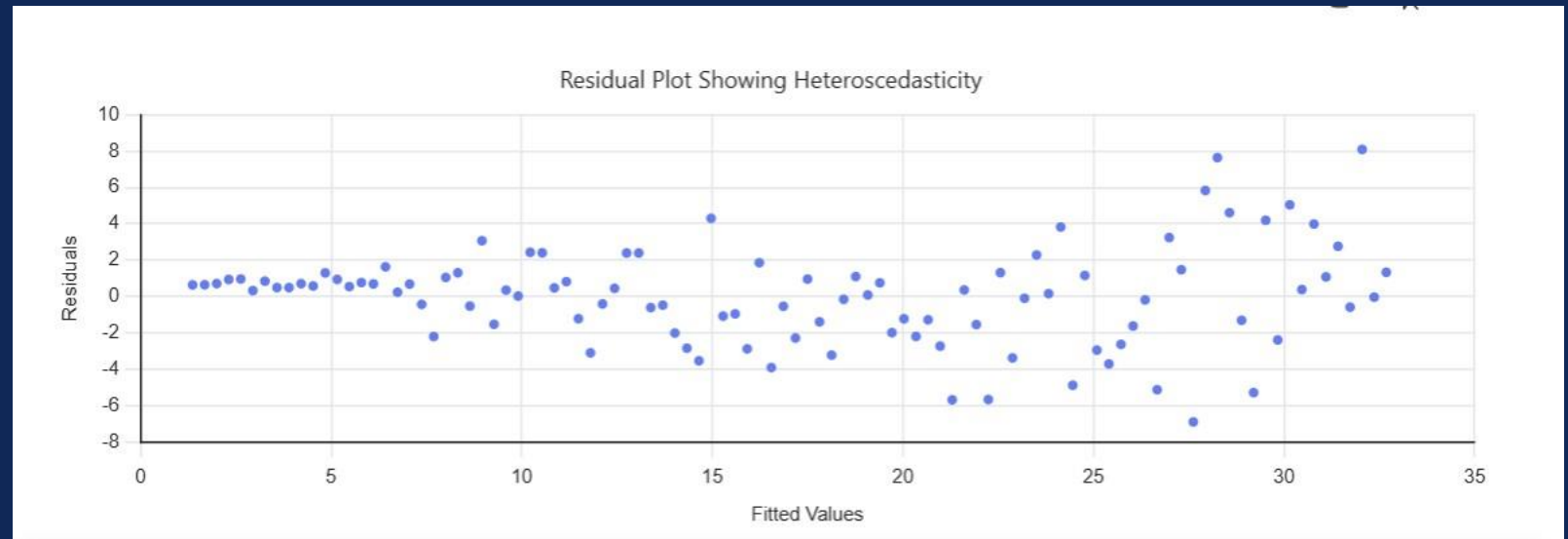
'Bad' Plot

ASSESS ASSUMPTIONS

- Independence: Each observation is independent from the other
- Example: One observation per person
- If multiple observations per person, need to use a different analyses
 - Mixed effects
 - Generalized estimating equations
 - Etc.
- If in time series further things to consider, beyond the scope of this module.

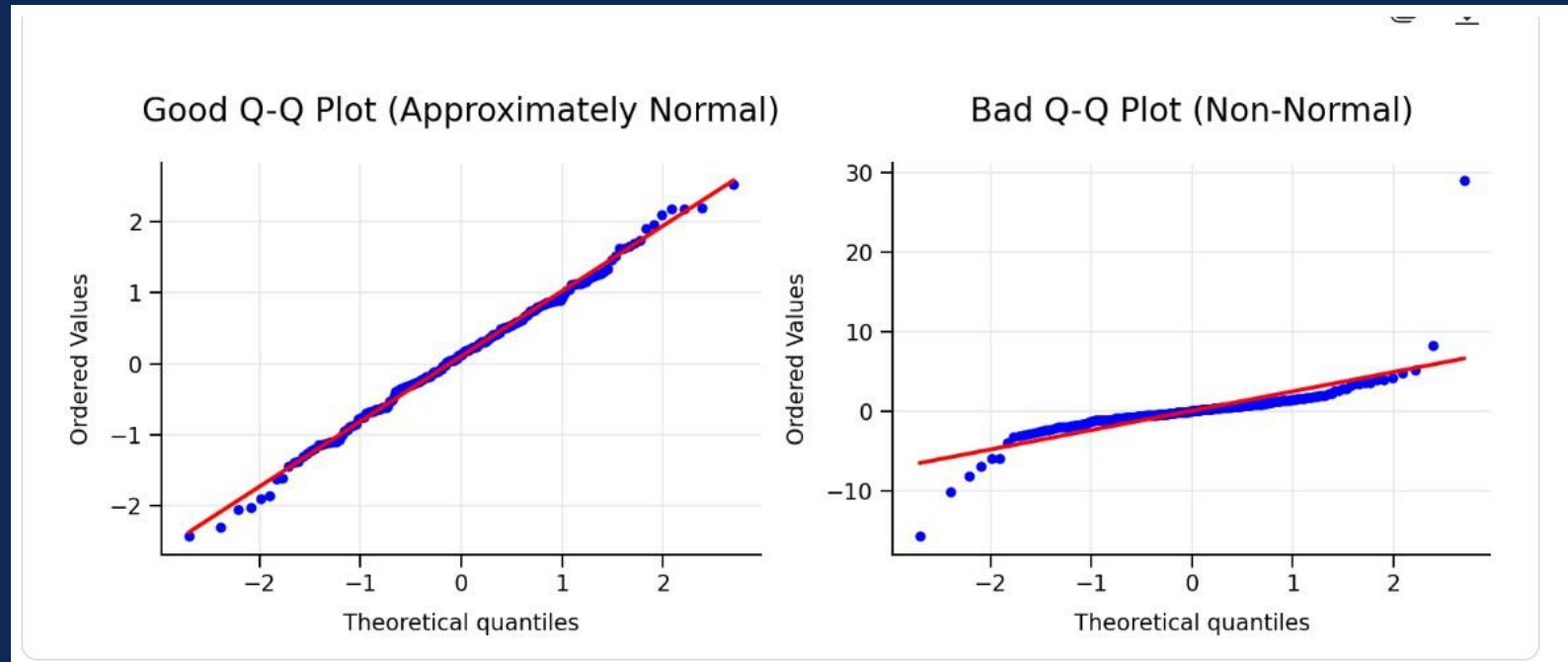
ASSESS ASSUMPTIONS

- Homoscedasticity: Constant variance between each point.
- This plot is showing how variance increases as scale increases
- Known as heteroscedasticity



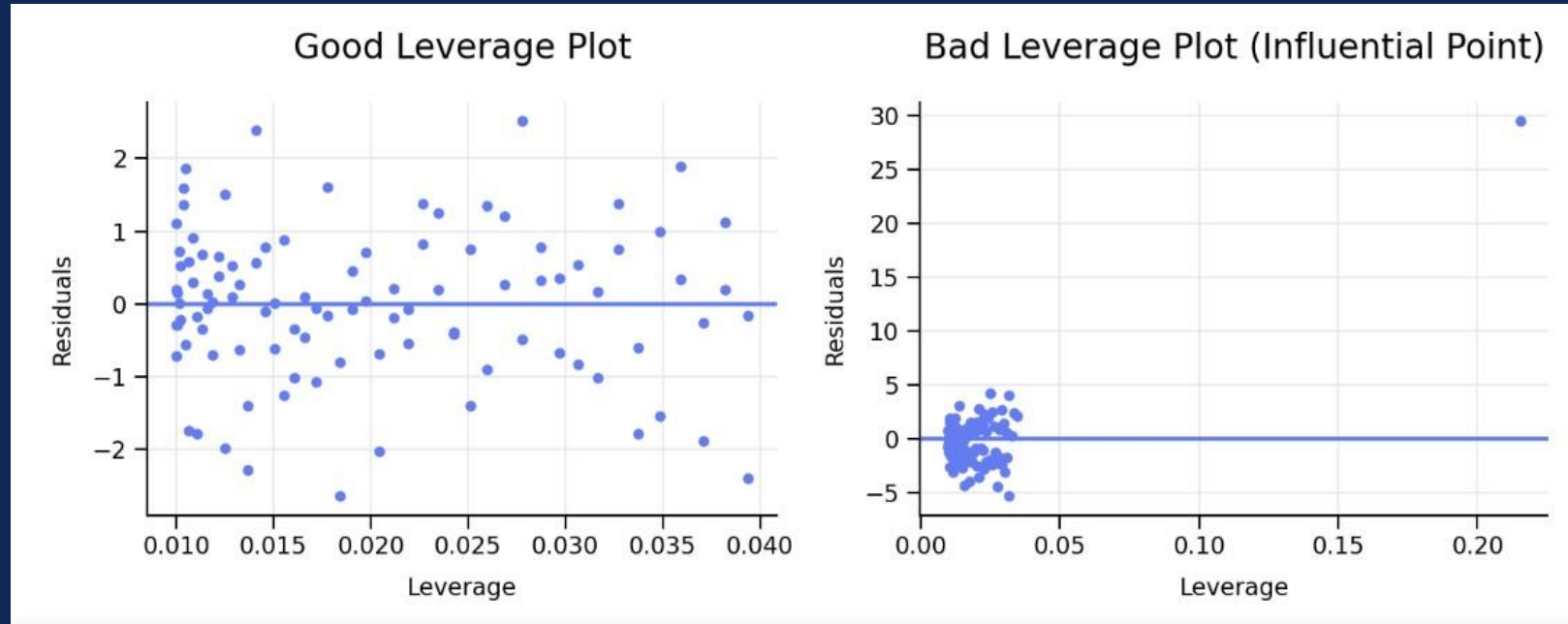
ASSESS ASSUMPTIONS

- Normality: That the residuals follow a normal distribution (bell curve)
- REMEMBER: This is residuals not raw data!

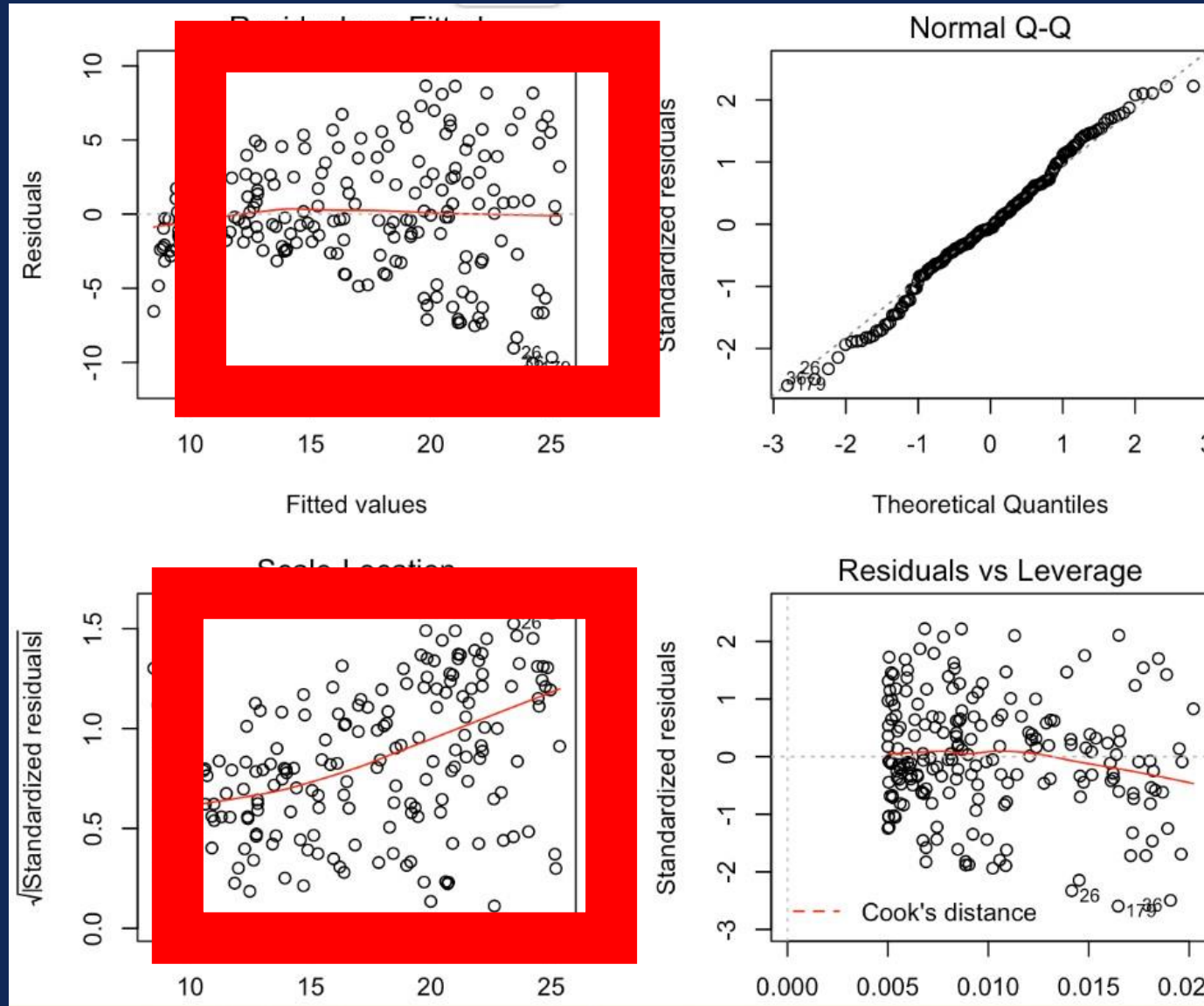


ASSESS ASSUMPTIONS

- Leverage: The 'weight' of one point and how this effects the overall regression



ASSESS ASSUMPTIONS



WHAT COULD WE DO ABOUT THIS?

- Log transformation `model <- glm(Y ~ log(X), data = data, family = 'gaussian')`
- Robust standard errors: 'Sandwich' estimator, beyond the scope of this module
- Assess outliers
 - Can be important, and good practice, particularly with 'widespread' of data
 - Usually use 2.5 standard deviations beyond mean.
 - Exclude outliers and reassess

OK, SO WE GOT THE BASICS

- Example of an actual linear regression
- Will use the 'cars' dataset from base R.
- Use classically across R tutorials.
- Fully open source and already embedded into R!

```
#Load Data  
data(cars)
```

```
#View Data  
View(cars)  
head(cars)  
str(cars)
```

OK, SO WE GOT THE BASICS

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```
#Load Data  
data(cars)
```

```
#View Data  
View(cars)  
head(cars)  
str(cars)
```

	speed	dist
1	4	2
2	4	10
3	7	4
4	7	22
5	8	16
6	9	10
7	10	18
8	10	26
9	10	34
10	11	17
11	11	28
12	12	14
13	12	20

Showing 1 to 13 of 50 entries, 2 total columns

```
'data.frame':   50 obs. of  2 variables:  
 $ speed: num  4 4 7 7 8 9 10 10 10 11 ...  
 $ dist : num  2 10 4 22 16 10 18 26 34 17 ...
```

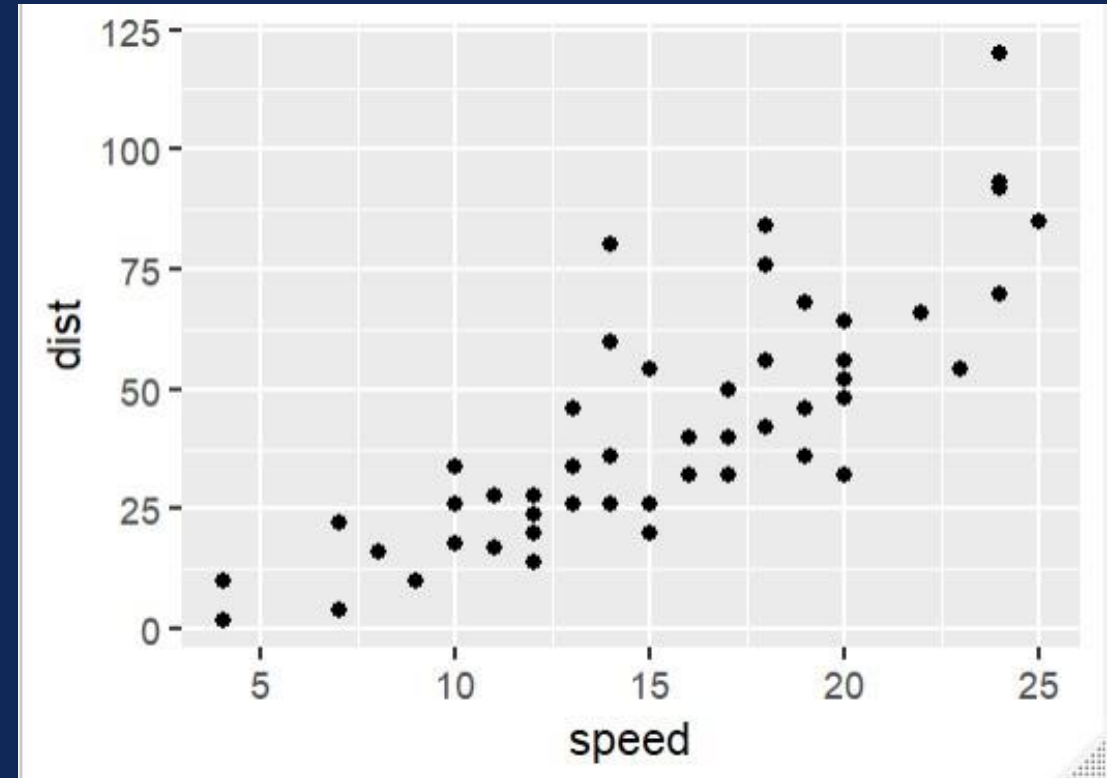
VIEW THE SCATTERPLOT

- Back to our Data Viz lecture!

```
library(ggplot2)
```

```
#Visualize Scatterplot
```

```
ggplot(cars, aes(x = speed, y = dist)) +  
geom_point()
```



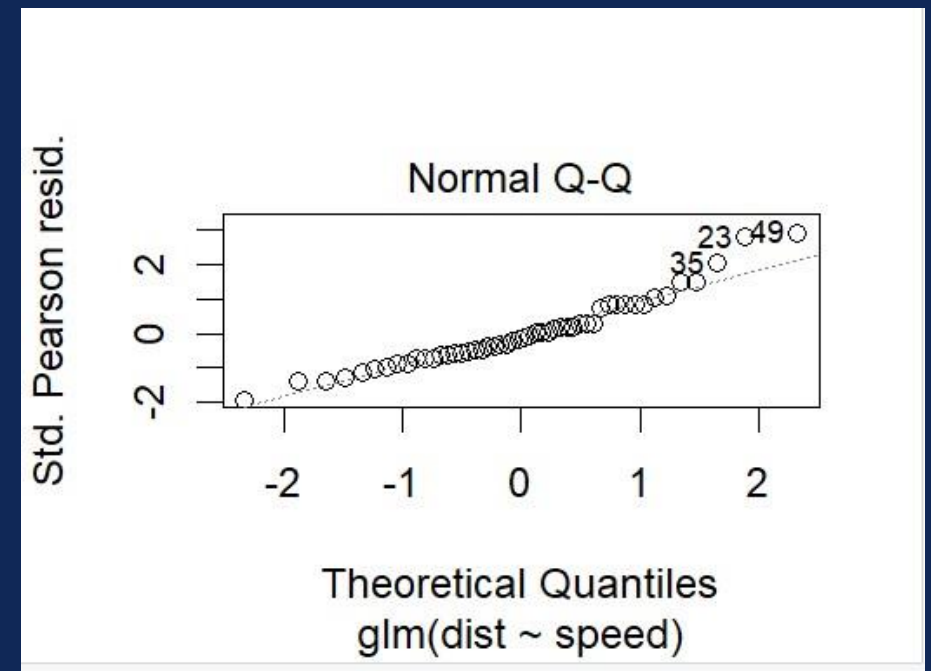
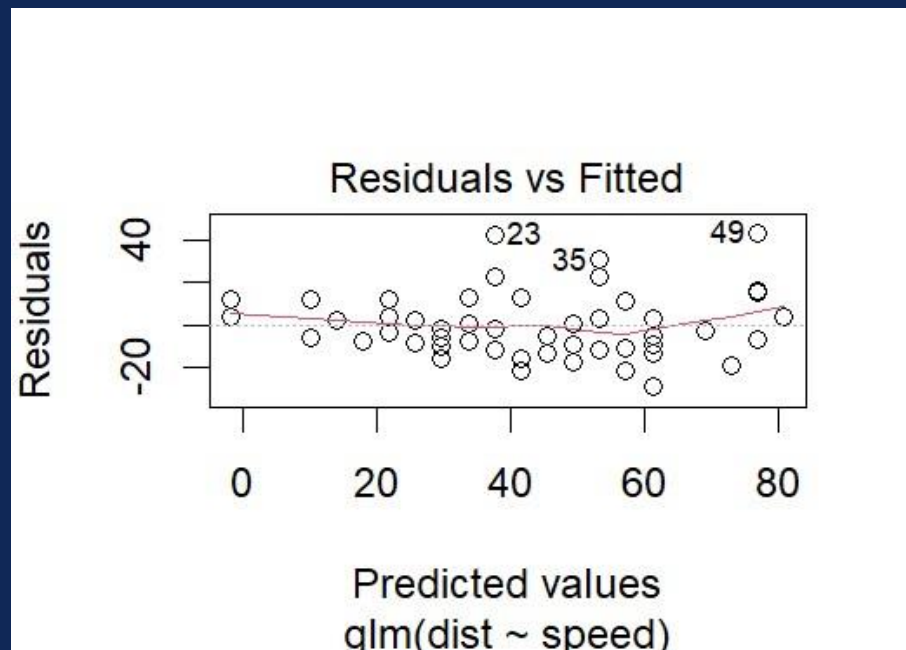
REGRESSION DIAGNOSTICS

#Regression of Association of Speed on Distance

```
model <- glm(dist ~ speed, data = cars, family = 'gaussian')
```

```
plot(model)
```

*Will see this: Hit <Return> To See Next Plot:



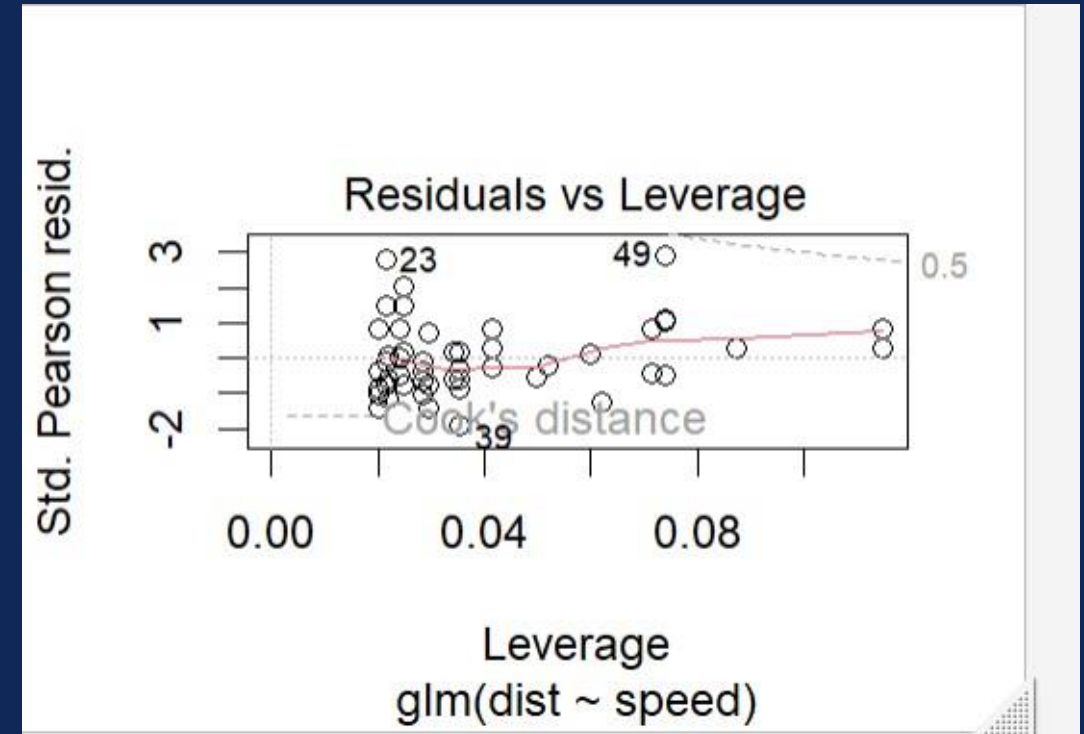
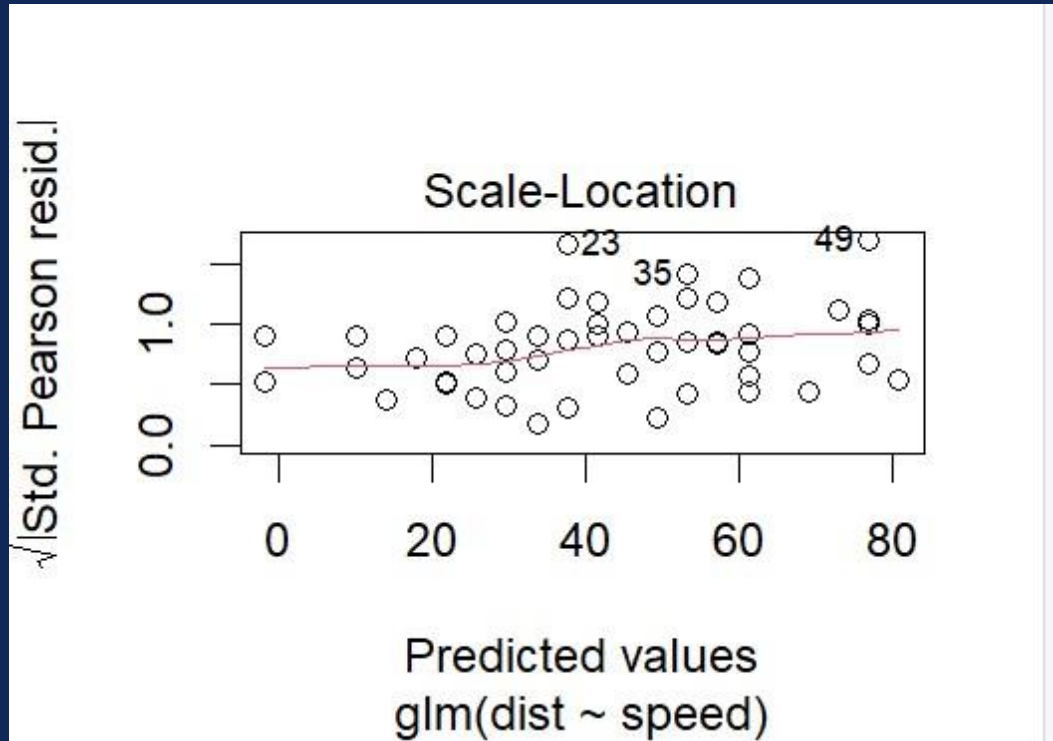
REGRESSION DIAGNOSTICS

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```
model <- glm(dist ~ speed, data = cars, family = 'gaussian')
```

```
plot(model)
```

*Will see this: Hit <Return> To See Next Plot:



PERFORM THE REGRESSION

```
library(Epi)
#Regression of Association of Speed on Distance
model <- glm(dist ~ speed, data = cars, family = 'gaussian')
ci.lin(model)
```

	Estimate	StdErr	z	P	2.5%	97.5%
(Intercept)	-17.579095	6.7584402	-2.601058	9.293674e-03	-30.825394	-4.332796
speed	3.932409	0.4155128	9.463990	2.964117e-21	3.118019	4.746799

MULTIPLE REGRESSION

- Adding in covariates to your model
- Remember, should be based on your scientific model
- Refer back to the DAG module

```
Model <- glm(Y ~ X1 + X2 + X3, data = data, family = 'gaussian')
```


MULTIPLE REGRESSION

Association & Causality: DO NOT INTERPRET 'other' covariates

- These variables are here to:
 - Control for variance
 - Block causal paths
- They are uninterpretable
- VERY IMPORTANT

JOURNAL ARTICLE

The Table 2 Fallacy: Presenting and Interpreting Confounder and Modifier Coefficients [Get access >](#)

Daniel Westreich ✉, Sander Greenland

American Journal of Epidemiology, Volume 177, Issue 4, 15 February 2013,
Pages 292–298, <https://doi.org/10.1093/aje/kws412>

Published: 30 January 2013 **Article history** ▼

MULTIPLE REGRESSION: ADDED ASSUMPTION

- No multicollinearity: When two or more variables are highly correlated.
- This can cause 'strange' results for the highly co-linear variables
- Examples:
 - Diastolic and Systolic Blood Pressure
 - Square meters of a house versus the number of rooms
 - 'Dummy Variable Trap': Single (0/1) v Married (0/1)

MULTIPLE REGRESSION: HOW DO WE ASSESS?

- Variance Inflation Factor (VIF): Measures severity of correlation.
 - How much variance inflation is due to correlation (hence VIF name 😊)
- `library(cars)`
- `model <- glm(Y ~ X1 + X2 + X3, data = data, family = 'gaussian')`
- `vif(model)`
- Correlation matrix
- `cor_matrix <- cor(data[, c("X1", "X2", "X3")])`
- `corrplot::corrplot(cor_matrix, method = "number")`

MULTIPLE REGRESSION: HOW DO WE ASSESS?

- Variance Inflation Factor (VIF):
 - `vif(model)`
 - $VIF > 10$
- Correlation matrix
 - `corrplot::corrplot(cor_matrix, method = "number")`
 - Correlation ≥ 0.80

MULTIPLE REGRESSION: ONE MORE THING

- Multicollinearity
 - Association: Can be important when deciding what to include for other covariates to control for.
 - Causality: We focus solely on exposure. Multicollinearity is much less important here
 - Prediction: This can be important, particularly for reducing parameters

LOGISTIC REGRESSION

- Binary outcome
- On the log odds scale
- Must exponentiate to get Odds Ratios (OR)
- `library(Epi)`
- `model <- glm(Y ~ X, data = data, family = 'binomial')`
- `ci.lin(model, Exp = T)`

LOGISTIC REGRESSION INTERPRETATION

- Odds Ratios start at 1.
- 1 = No association/effect
- Below 1 = Less Odds
- Above 1 = Greater odds

LOGISTIC REGRESSION ASSUMPTIONS

- Linearity of the log odds
- Independence of observations
- No complete quasi separation: Almost all 0's or all 1's

*These are not really about diagnostic plots but thinking through your data and questions.

FINAL THOUGHTS

- Regression is the tool you will use the most in analyses
- Know your outcome and exposure – this will guide your analytical strategy
- Your question will guide your inference
- Regression diagnostics are a dark art